patterns. More abstract geometric ideas are also introduced at this Stage and relations with algebra are explored to solve problems and puzzles.

Finally, the Secondary Stage focusses on further developing the ability to justify claims and arguments through logical reasoning. Students become comfortable in working with abstractions and other core techniques of Mathematics and Computational Thinking, such as the mathematical modelling of phenomena and the development of algorithms to solve problems.

Across the Stages, students develop mathematical skills such as problem solving, visualisation, optimisation, representation, and communication, and thereby developing the capacities of Mathematics and Computational Thinking. Through creating and solving puzzles, pictorials, word problems, and optimisation problems, various values and dispositions, such as, perseverance, curiosity, confidence, rigour, and honesty would be developed across grades.

Finally, Mathematics has an extremely rich history in India spanning thousands of years. India is where the place value number system (including zero), that we all use today to write numbers, was first developed and used and is where many of the key foundations of algebra, geometry, trigonometry, and calculus were laid. By learning about the development of Mathematics in India as well as throughout the world, the rootedness in India can be enhanced, along with a more general appreciation of the history of Mathematics, and of the remarkable evolution and development of mathematical concepts through time (and India's critical roles in these developments).

Curricular Goals and Competencies for Preparatory Stage

Curricular Goals	Competencies
CG-1 Understands numbers (counting numbers and fractions), represents whole numbers using the Indian place value system, understands and carries out the four basic operations with whole numbers, and discovers and recognises patterns in number sequences	1.1 Represents numbers using the place value structure of the Indian number system, compares whole numbers, and knows and can read the names of very large numbers
	1.2 Represents and compares commonly used fractions in daily life (such as ½, ¼) as parts of unit wholes, as locations on number lines and as divisions of whole numbers
	1.3 Understands and visualises arithmetic operations and the relationships among them, knows addition and multiplication tables at least up to 10x10 (pahade) and applies the four basic operations on whole numbers to solve daily life problems

	1.4 Recognises, describes, and extends simple number patterns such as odd numbers, even numbers, square numbers, cubes, powers of 2, powers of 10, and Virahanka–Fibonacci numbers.
CG-2 Analyses the characteristics and properties of two - and three-dimensional geometric shapes, specifies locations and describes spatial relationships, and recognises and creates shapes that have symmetry	2.1 Identifies, compares, and analyses attributes of two- and three-dimensional shapes and develops vocabulary to describe their attributes/properties
	2.2 Describes location and movement using both common language and mathematical vocabulary; understands the notion of map (najri naksha)
	2.3 Recognises and creates symmetry (reflection, rotation) in familiar 2D and 3D shapes
	2.4 Discovers, recognises, describes, and extends patterns in 2D and 3D shapes
CG-3 Understands measurable attributes of objects and the units, systems, and processes of such measurement, including those related to distance, length, weight, area, volume, and time using nonstandard and standard units	3.1 Measures in non-standard and standard units and evaluates the need for standard units
	3.2 Uses an appropriate unit and tool for the attribute (like length, perimeter, time, weight, volume) being measured
	3.3 Carries out simple unit conversions, such as from centimetres to metres, within a system of measurement
	3.4 Understands the definition and formula for the area of a square or rectangle as length times breadth
	3.5 Devises strategies for estimating the distance, length, time, perimeter (for regular and irregular shapes), area (for regular and irregular shapes), weight, and volume and verifies the same using standard units
	3.6 Deduces that shapes having equal areas can have different perimeters and shapes having equal perimeters can have different areas
	3.7 Evaluates the conservation of attributes like length and volume, and solves daily-life problems related to them

CG-4 Develops problem-solving skills with procedural fluency to solve mathematical puzzles as well as daily-life problems, and as a step towards developing computational thinking	4.1 Solves puzzles and daily-life problems involving one or more operations on whole numbers (including word puzzles and puzzles from 'recreational' areas, such as the construction of magic squares)
	4.2 Learns to systematically count and list all possible permutations or combination given a constraint, in simple situations (e.g., how to make a committee of two people from a group of five people)
	4.3 Selects appropriate methods and tools for computing with whole numbers, such as mental computation, estimation, or paperpencil calculation, in accordance with the context
CG-5 Knows and appreciates the development in India of the decimal place value system that is used around the world today	5.1 Understands the development of zero in India and the Indian place value system for writing numerals, the history of its transmission to the world, and its modern impact on our lives and in all technology

Principles for Content Selection

The following principles will be followed while choosing topics of study for Mathematics classrooms. Stage-wise principles are laid down; for each stage, principles for the previous Stage have also been considered, wherever applicable.

Preparatory Stage

a. Plenty of space should be given to students' local context and surroundings for developing concepts in Mathematics. Case studies, stories, situations from daily life, and vocabulary and phrasing in the home language should be brought in to help introduce and unfold a concept and its sub-concepts.

b. The development of a culture of learning outside the classroom should be encouraged. More play-way methods (activities) should be included wherever possible.

c. Avenues for mathematical reasoning should be created in all activities, projects, assignments, and exercises. The content should encourage students to articulate the reasons behind their observations and guesses or conjectures and ask why a pattern extends in a certain way and what the rule behind it is.

d. The language of the content should be simple so that students can also express their thoughts using similar language; this should gradually enlarge their vocabulary and enable them to become more skilled over time in using precise mathematical vocabulary, symbols, and notation.